

SIH PROJECT — INTEGRATED MASTER BLUEPRINT

Project: AI-Powered Freight Forecasting, Vessel Chartering & Bulk Cargo Procurement Decision System

Scope: Forecasting, Vessel Selection, Port Feasibility, Route Optimization, Transshipment/Lightering, Taxes/Tariffs, Canal Costs, Total Landed Cost, Risk and Procurement Timing.

1. Final Project Objective

Given cargo type, quantity, origin, destination, delivery deadline, budget and risk tolerance, the system recommends what and when to procure, which vessel to charter, which route and port strategy to use, whether direct shipment or transshipment is preferable, and the expected total landed cost.

2. Final Data Architecture — 10 Extended Tables

The database architecture maps all logistical, financial, infrastructure, and compliance dimensions needed for strict regulatory and dimensional feasibility checking.

1. PORT_MASTER

port_id	port_name
country	port_type
channel_depth_m	berth_depth_m
max_draft_m	max_loa_m
max_beam_m	max_vessel_dwt
tidal_restriction	tidal_window_hours
berth_count	berth_availability_pct
loading_rate_mt_day	discharge_rate_mt_day
port_handling_cost_usd_mt	average_waiting_time_days
congestion_index	weather_risk_index
requires_shore_cranes	[ADDED] boolean indicator for gearless vessel filtering
port_dues_currency	[ADDED] currency code for localized Scale of Rates handling

2. VESSEL_MASTER

vessel_id	vessel_type
dwt	capacity_mt
loa_m	beam_m

draft_m	speed_knots
fuel_consumption_mt_day	port_fuel_consumption_mt_day
age_years	vessel_status
availability_date	current_position
current_port	charter_rate_usd_day
has_onboard_cranes	[ADDED] boolean flag to match ports requiring geared ships
cii_rating	[ADDED] carbon efficiency rating (A-E) for regulatory tracking
fuel_type	[ADDED] primary fuel compliance metric (VLSFO / LSMGO / LNG)

3. VESSEL_PORT_COMPATIBILITY

vessel_id	port_id
draft_feasible	loa_feasible
beam_feasible	dwt_feasible
tidal_feasible	port_access_score
access_status	restriction_reason
gear_requirement_met	[ADDED] flags crane mismatches at unequipped terminals

4. MARKET_DATA

date	origin_region
destination_region	cargo_type
freight_rate_usd_mt	bunker_price_usd_t
commodity_price_usd_t	commodity_demand_index
trade_volume_mt	vessel_availability_index
fleet_utilization_pct	fleet_orderbook_pct
port_congestion_index	weather_risk_index
geopolitical_risk_index	usd_inr
season	historical_freight_volatility
exchange_rate_date	[ADDED] temporal marker for INR/USD volatility mapping

5. VOYAGE_DATA

voyage_id	date
vessel_id	origin_port
destination_port	cargo_type
cargo_quantity_mt	freight_rate_usd_mt
charter_rate_usd_day	distance_nm
voyage_duration_days	loading_time_days
discharge_time_days	waiting_time_days
port_calls	transshipment_required
lighterage_required	actual_total_cost_usd
actual_cost_usd_mt	
deadfreight_charge_usd	[ADDED] penalty fee for under-utilized charter holds
yard_detention_cost_usd	[ADDED] penalty for slow bulk evacuation from port side

6. VOYAGE_LEGS

voyage_id	leg_number
from_port_id	to_port_id
vessel_id	operation
cargo_loaded_mt	cargo_discharged_mt
distance_nm	sailing_days
fuel_required_mt	fuel_cost_usd
port_time_days	port_cost_usd
leg_cost_usd	

7. TRANSSHIPMENT_DATA

transshipment_id	voyage_id
mother_vessel_id	transshipment_port_id
feeder_vessel_id	transshipment_required
cargo_transferred_mt	transshipment_time_days
transshipment_cost_usd_mt	lighterage_required
lighterage_capacity_mt	lighterage_distance_nm
lighterage_cost_usd_mt	lighterage_time_days
sandheads_lighterage_trigger	[ADDED] strict binary flag for Kolkata/Haldia riverine channel entries

8. ROUTE_DATA

route_id	origin_port_id
-----------------	----------------

destination_port_id	distance_nm
canal_used	canal_name
canal_transit_cost_usd	canal_transit_days
canal_surcharge_usd	piracy_risk_index
weather_risk_index	geopolitical_risk_index
route_disruption_probability	seasonal_delay_days

9. CARGO_REQUEST

request_id	cargo_type
cargo_quantity_mt	cargo_purchase_price_usd_mt
origin_port	destination_port
earliest_loading_date	required_delivery_date
maximum_budget_usd_mt	maximum_risk_score
preferred_vessel_type	allow_transshipment
allow_cargo_split	allow_lighterage

10. TAX_TARIFF_DATA

tariff_id	origin_country
destination_country	cargo_type
hs_code	customs_duty_pct
import_tariff_pct	export_duty_pct
igst_pct	cess_pct
other_tax_pct	tariff_rate_usd_mt
effective_from	effective_to
trade_agreement	

3. Final Calculated Variables

- **Transportation:** sea_distance_nm, sailing_days, fuel_required_mt, fuel_cost_usd, port_time_days, port_cost_usd, waiting_cost_usd, vessel_operating_cost_usd, carbon_intensity_emissions_mt [ADDED]
- **Transshipment:** transshipment_cost_usd, transshipment_time_days, lighterage_cost_usd, lighterage_time_days
- **Trade:** customs_duty_usd, import_tariff_usd, import_tax_usd, igst_usd, cess_usd, other_tax_usd, forex_volatility_buffer [ADDED]
- **Route:** canal_cost_usd, canal_time_days, delay_cost_usd, risk_penalty_usd
- **Final:** cargo_purchase_cost_usd, total_logistics_cost_usd, total_landed_cost_usd, landed_cost_usd_mt, deadfreight_penalty_usd [ADDED], yard_detention_cost_usd [ADDED]

4. Upgraded Total Landed Cost Equation

TOTAL LANDED COST = Cargo Purchase Cost + Ocean Freight + Charter Cost + Fuel Cost + Port Charges + Waiting/Demurrage + Transshipment + Lighterage + Canal Charges + Customs Duty + Import Tariff + Applicable Taxes + Handling Charges + Delay Cost + Risk Penalty + **Deadfreight Penalty [ADDED]** + **Yard Detention Fees [ADDED]**

LANDED COST / MT = TOTAL LANDED COST ÷ CARGO QUANTITY

5. Final ML Model & Time-Series Rigour

Inputs: historical freight, bunker price, commodity price, demand, trade volume, vessel availability, fleet utilization, port congestion, weather risk, geopolitical risk, USD/INR, seasonality, lag features and moving averages.

Models: Linear Regression baseline → Random Forest → Gradient Boosting/XGBoost / LSTM.

Metrics: MAE, RMSE, MAPE, R².

Execution Guardrail [ADDED]: To completely prevent data leakage inherent in temporal features, the system strictly enforces a rolling-window *TimeSeriesSplit* instead of randomized K-Fold validation. Explanations are served globally and locally via SHAP values directly integrated inside the analytical dashboard view.

6. Final Optimization Engine (Google OR-Tools)

Decision variables: vessel, vessel size, route, origin port, destination port, transshipment port, direct/transshipment choice, number of port calls, cargo allocation, feeder/lighter, chartering timing and procurement timing.

Operational Constraints: cargo requirement, vessel capacity, port draft, vessel LOA, beam, DWT, tidal restrictions, vessel availability, port capacity, delivery deadline, risk limit and budget.

- **Laycan Overlap Constraint [ADDED]:** Blocks vessel scheduling logic if the preceding voyage's estimated discharge completion window overlaps with the target loading window.
- **Infrastructure Compatibility Constraint [ADDED]:** Automatically cross-checks port crane equipment against vessel gearing profiles to eliminate unloads failures.

7. Final Scenario Engine & East Coast India Realities

- **Scenario A — Direct:** Deep routes, e.g., Indonesia / Australia → Dhamra or Gangavaram (Capesize compliant).
- **Scenario B — Transshipment/Lighterage:** Automatically triggered at hubs like Sandheads for riverine entry into Kolkata/Haldia docks to avoid grounding shallow berths.
- **Scenario C — Cargo Split:** Cargo volumes divided dynamically across multiple smaller sister vessels to fit narrow port schedules or seasonal monsoon constraints.

8. Final Risk Engine

Calculates weather risk (including Bay of Bengal seasonal cyclonic patterns from October to December), geopolitical friction indices, canal choke-point restrictions, port congestion, and vessel availability queues. Yields an overall 0–100 risk score paired with explicit LOW/MEDIUM/HIGH risk category assignments.

9. Final Procurement Intelligence

The system translates raw forecasting signals into structural tactical directives. By analyzing current charter indicators against forward ML predictions, the module issues unambiguous ****Book Now**** or ****Wait**** signals to lock optimal financial positions.

10. Final Analytical Dashboard Layout

- **Cargo Request:** Handles interactive inputs for commodity profiles, volumes, and time constraints.
- **Freight Forecast:** Displays 7/30-day directional outputs backed by interactive SHAP driver breakdowns.
- **Vessel & Route Engine:** Parallel side-by-side matrices contrasting Direct vs. Lighterage costs and physical limits.
- **Landed Cost Component:** Highly granulated financial itemization ending in a definitive Landed Cost per Metric Ton.

11. Final Technical Stack

Data Layer: Python, Pandas, NumPy, PostgreSQL

Machine Learning: scikit-learn, XGBoost, SHAP, Prophet

Optimization Framework: Google OR-Tools (Mixed-Integer Linear Programming)

Backend Controllers: FastAPI, Pydantic robust validation schemas

Frontend UI Experience: React.js, TailwindCSS, Chart.js, Leaflet Maps for interactive vessel tracks

12. Final Team Responsibilities & Implementation Guardrails

- **Member 1 (Lead + Backend):** Core API routing, optimization controller loops, infrastructure mapping.
- **Member 2 (Data Engineer):** ETL processing pipelines, Indian East Coast localized port tariff matrices ingestion.

- **Member 3 (ML Engineer):** Temporal feature generation, TimeSeriesSplit execution, SHAP value extractions.
- **Member 4 (Optimization Engineer):** OR-Tools implementation, lighterage boundary calculations, multi-port legs.
- **Member 5 (Frontend Engineer):** Map visual plotting, scenario comparison matrix rendering, procurement alerts UI.